Problem Set for Linear Programming

1. Solve the following LPs graphically.

   (a)

$$\begin{aligned} \max \quad & x_1 + x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 \le 6, \\ & x_1 \le 4, \\ & x_1, x_2 \ge 0. \end{aligned}$$

   (b)

$$\begin{aligned} \max \quad & x_1 - x_2 \\ \text{s.t.} \quad & x_1 + x_2 \le 5, \\ & x_1 - x_2 \ge 1, \\ & x_1, x_2 \ge 0. \end{aligned}$$

   (c)

$$\begin{aligned} \max \quad & -x_1 + 2x_2 \\ \text{s.t.} \quad & x_1 + x_2 \le 4, \\ & x_1 \ge 1, \\ & x_2 \le 3, \\ & x_1, x_2 \ge 0. \end{aligned}$$

   (d)

$$\begin{aligned} \max \quad & -3x_1 - 2x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \ge 4, \\ & x_1 + 3x_2 \ge 6, \\ & x_1 \le 5, \\ & x_2 \le 5, \\ & x_1, x_2 \ge 0. \end{aligned}$$

2. Write the dual form of the following two LPs.

   (a)

$$\begin{aligned} \min \ & 2x_1 + 2x_2 + 4x_3 \\ & 2x_1 + 3x_2 + 5x_3 \ge 2 \\ & 3x_1 + x_2 + 7x_3 \le 3 \\ & x_1 + 4x_2 + 6x_3 \le 5 \\ & x_1, x_2, x_3 \ge 0 \end{aligned}$$

   (b)

$$\begin{aligned} \max \ & x_1 + 2x_2 + 3x_3 + 4x_4 \\ & -x_1 + x_2 - x_3 - 3x_4 = 5 \\ & 6x_1 + 7x_2 + 3x_3 - 5x_4 \ge 8 \\ & 12x_1 - 9x_2 - 9x_3 + 9x_4 \le 20 \\ & x_1 \ge 0, x_2 \ge 0, \ x_3 \le 0, \ x_4 \text{ free} \end{aligned}$$

3. The Primo Insurance Company is introducing two new product lines: special risk insurance and mortgages. The expected profit is \$5 per unit on special risk insurance and \$2 per unit on mortgages.

   Management wishes to establish sales quotas for the new product lines to maximize total expected profit. The work requirements are as follows:

Department	**Work-Hours per Unit**		
	Special Risk	**Mortgage**	**Work-Hours Available**
Underwriting	3	2	2400
Administration	0	1	800
Claims	2	0	1200

   (a) Formulate a linear programming model for this problem.

   (b) Use the graphical method to solve this model.

4. The SOUTHERN CONFEDERATION OF KIBBUTZIM is a group of three kibbutzim (communal farming communities) in Israel. Overall planning for this group is done in its Coordinating Technical Office. This office currently is planning agricultural production for the coming year.

   The agricultural output of each kibbutz is limited by both the amount of available irrigable land and the quantity of water allocated for irrigation by the Water Commissioner (a national government official). These data are given in the following table.

Kibbutz	Usable Land (Acres)	Water Allocation (Acre Feet)
1	400	600
2	600	800
3	300	375

   The crops suited for this region include sugar beets, cotton, and sorghum, and these are the three being considered for the upcoming season. These crops differ primarily in their expected net return per acre and their consumption of water. In addition, the Ministry of Agriculture has set a maximum quota for the total acreage that can be devoted to each of these crops by the Southern Confederation of Kibbutzim, as shown in the second table below.

Crop	Maximum Quota (Acres)	Water Consumption (Acre Feet/Acre)	Net Return (\$/Acre)
Sugar beets	600	3	1000
Cotton	500	2	750
Sorghum	325	1	250

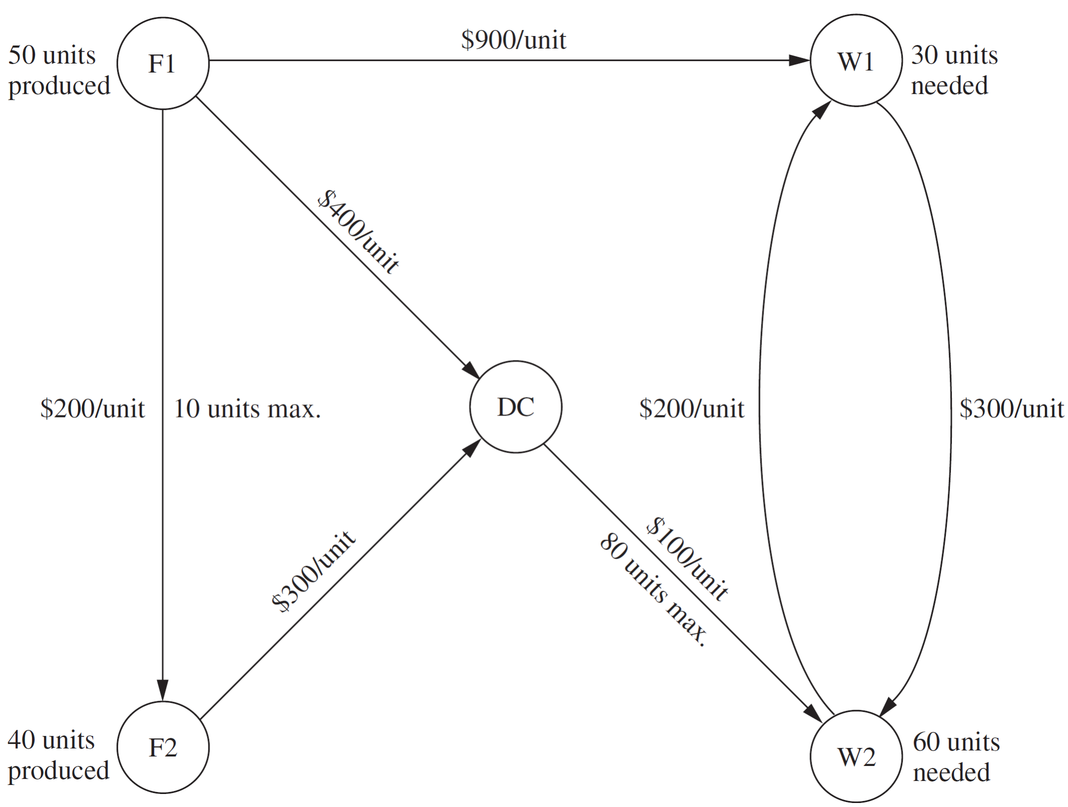
   Because of the limited water available for irrigation, the Southern Confederation of Kibbutzim will not be able to use all its irrigable land for planting crops in the upcoming season. To ensure equity between the three kibbutzim, it has been agreed that every kibbutz will plant the same proportion of its available irrigable land. For example, if kibbutz 1 plants 200 of its available 400 acres, then kibbutz 2 must plant 300 of its 600 acres, while kibbutz 3 plants 150 acres of its 300 acres. However, any combination of the crops may be grown at any of the kibbutzim. The job facing the Coordinating Technical Office is to plan how many acres to devote to each crop at the respective kibbutzim while satisfying the given restrictions. The objective is to maximize the total net return to the Southern Confederation of Kibbutzim as a whole.

   Suppose the quantities to be decided upon are the number of acres to devote to each of the three crops at each of the three kibbutzim. The decision variables x_j for $j = 1, 2, ..., 9$ represent these nine quantities, as shown in the following table. Write down the LP model for this problem.

Crop	Allocation (Acres)		
	Kibbutz		
	1	2	3
Sugar beets	x_1	x_2	x_3
Cotton	x_4	x_5	x_6
Sorghum	x_7	x_8	x_9

5. The DISTRIBUTION UNLIMITED CO. will be producing the same new product at two different factories, and then the product must be shipped to two warehouses, where either factory can supply either warehouse. The distribution network available for shipping this product is shown in the figure below, where $F1$ and $F2$ are the two factories, $W1$ and $W2$ are the two warehouses, and DC is a distribution center. The amounts to be shipped from $F1$ and $F2$ are shown to their left, and the amounts to be received at $W1$ and $W2$ are shown to their right. Each arrow represents a feasible shipping lane. Thus, $F1$ can ship directly to $W1$ and has three possible routes ($F1 \to DC \to W2$, $F1 \to F2 \to DC \to W2$, and $F1 \to W1 \to W2$) for shipping to $W2$. Factory $F2$ has just one route to $W2$ ($F2 \to DC \to W2$) and one to $W1$ ($F2 \to DC \to W2 \to W1$). The cost per unit shipped through each shipping lane is shown next to the arrow. Also shown next to $F1 \to F2$ and $DC \to W2$ are the maximum amounts that can be shipped through these lanes. The other lanes have sufficient shipping capacity to handle everything these factories can send. The decision to be made concerns how much to ship through each shipping lane. The objective is to minimize the total shipping cost.

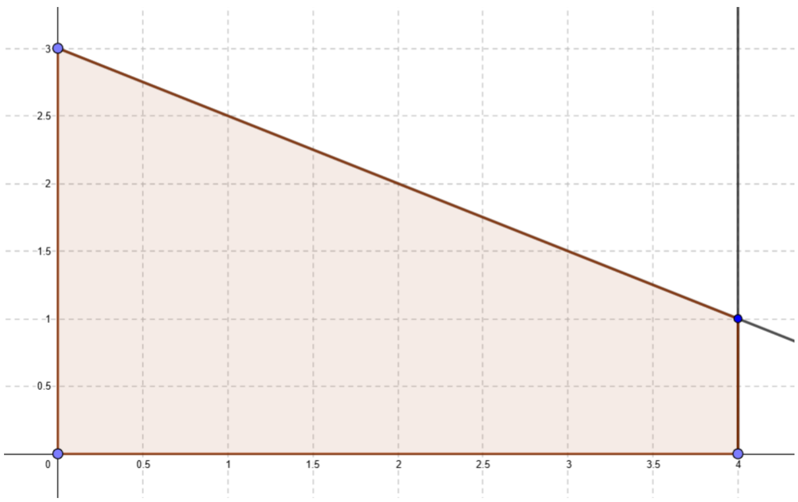
   With seven shipping lanes, we need seven decision variables (x_{F1-F2}, x_{F1-DC}, x_{F1-W1}, x_{F2-DC}, x_{DC-W2}, x_{W1-W2}, x_{W2-W1}) to represent the amounts shipped through the respective lanes. Write down the LP model for this problem.



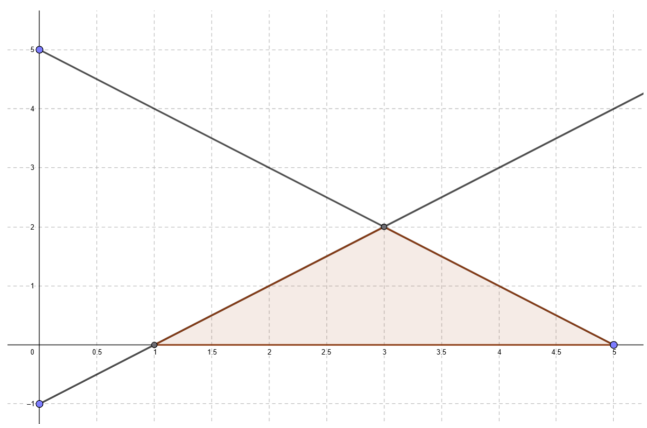
Solution

1. The optimal solutions are

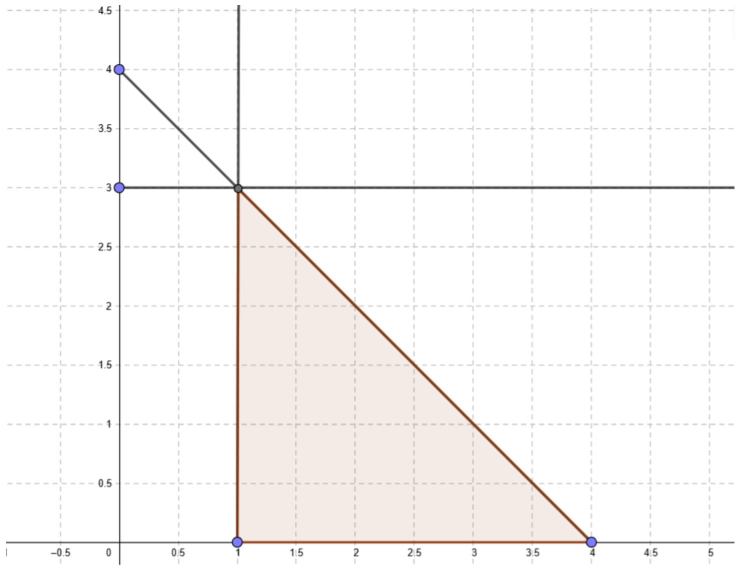
   (a) (4, 1)



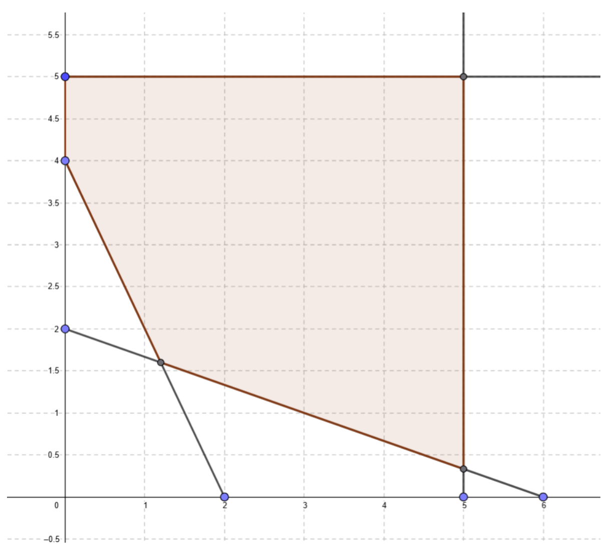
   (b) (5, 0)



   (c) (1, 3)



(d) (1.2, 1.6)



2. The dual problems are

   (a)

$$
\begin{aligned}
\max \quad & 2y_1 + 3y_2 + 5y_3 \\
\text{s.t.} \quad & 2y_1 + 3y_2 + y_3 \leq 2 \\
& 3y_1 + y_2 + 4y_3 \leq 2 \\
& 5y_1 + 7y_2 + 6y_3 \leq 4 \\
& y_1 \geq 0,\ y_2 \leq 0,\ y_3 \leq 0
\end{aligned}
$$

(b)

$$
\begin{aligned}
\min \quad & 5y_1 + 8y_2 + 20y_3 \\
\text{s.t.} \quad & -y_1 + 6y_2 + 12y_3 \geq 1 \\
& y_1 + 7y_2 - 9y_3 \geq 2 \\
& -y_1 + 3y_2 - 9y_3 \leq 3 \\
& -3y_1 - 5y_2 + 9y_3 = 4 \\
& y_1 \text{ free}, \ y_2 \leq 0, \ y_3 \geq 0
\end{aligned}
$$

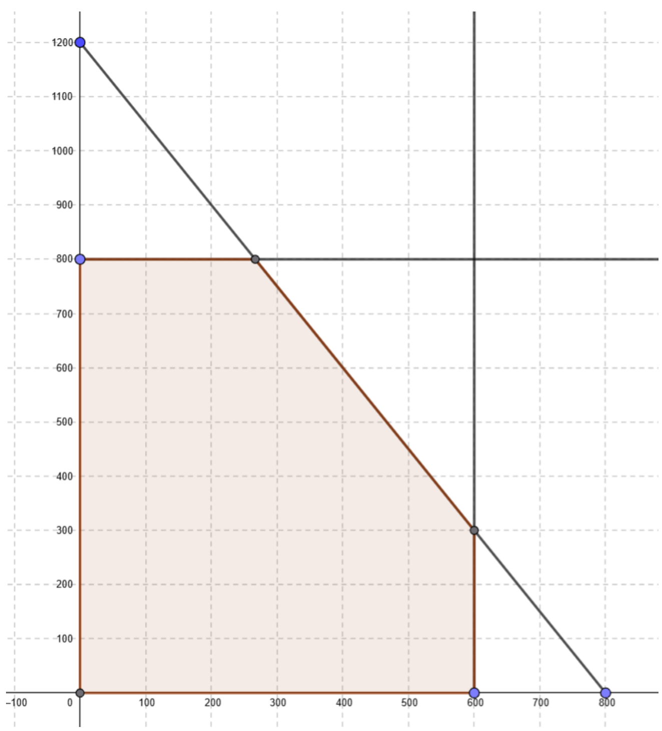
3. Let x_1 and x_2 be the numbers of units of special risk insurance and units of mortgages, respectively. Then we have

   (a) Objective Function (Maximize total profit): Maximize $5x_1 + 2x_2$

   (b) Constraints:

$$
\begin{aligned}
\text{Underwriting:} \quad & 3x_1 + 2x_2 \leq 2400 \\
\text{Administration:} \quad & x_2 \leq 800 \\
\text{Claims:} \quad & 2x_1 \leq 1200 \\
\text{Non-negativity:} \quad & x_1 \geq 0, \quad x_2 \geq 0
\end{aligned}
$$

   The feasible region is



   The optimal solution is (600, 300). The corresponding optimal objective function value is 3600.

4. From the problem description, the Coordinating Technical Office plants:

   - x_1, x_2, x_3 acres of sugar beets in Kibbutz 1, 2, 3 respectively
   - x_4, x_5, x_6 acres of cotton in Kibbutz 1, 2, 3 respectively
   - x_7, x_8, x_9 acres of sorghum in Kibbutz 1, 2, 3 respectively

It maximizes the total net return:

$$\text{Maximize } 1000(x_1 + x_2 + x_3) + 750(x_4 + x_5 + x_6) + 250(x_7 + x_8 + x_9)$$

while considering the following constraints.

(a) Land Constraints (by Kibbutz)

$$\begin{aligned} x_1 + x_4 + x_7 &\leq 400 \quad \text{(Kibbutz 1)} \\ x_2 + x_5 + x_8 &\leq 600 \quad \text{(Kibbutz 2)} \\ x_3 + x_6 + x_9 &\leq 300 \quad \text{(Kibbutz 3)} \end{aligned}$$

(b) Water Constraints (by Kibbutz)

$$\begin{aligned} 3x_1 + 2x_4 + x_7 &\leq 600 \quad \text{(Kibbutz 1)} \\ 3x_2 + 2x_5 + x_8 &\leq 800 \quad \text{(Kibbutz 2)} \\ 3x_3 + 2x_6 + x_9 &\leq 375 \quad \text{(Kibbutz 3)} \end{aligned}$$

(c) Crop Quota Constraints

$$\begin{aligned} x_1 + x_2 + x_3 &\leq 600 \quad \text{(Sugar beets)} \\ x_4 + x_5 + x_6 &\leq 500 \quad \text{(Cotton)} \\ x_7 + x_8 + x_9 &\leq 325 \quad \text{(Sorghum)} \end{aligned}$$

(d) Proportional Land Use Constraints

$$\begin{aligned} \frac{x_1 + x_4 + x_7}{400} &= \frac{x_2 + x_5 + x_8}{600} \\ \frac{x_1 + x_4 + x_7}{400} &= \frac{x_3 + x_6 + x_9}{300} \end{aligned}$$

which simplify to:

$$\begin{aligned} 3x_1 + 3x_4 + 3x_7 - 2x_2 - 2x_5 - 2x_8 &= 0 \\ 3x_1 + 3x_4 + 3x_7 - 4x_3 - 4x_6 - 4x_9 &= 0 \end{aligned}$$

(e) Non-negativity Constraints

$$x_j \geq 0, \quad j = 1, 2, \ldots, 9$$

The optimal solution is $(x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9) = (133.3333, 100, 25, 100, 250, 150, 0, 0, 0)$. You can try it using Excel.

5. Suppose we define the shipping amounts through each lane as follows:

- x_{F1-F2}: Units shipped from Factory 1 to Factory 2
- x_{F1-DC}: Units shipped from Factory 1 to Distribution Center
- x_{F1-W1}: Units shipped from Factory 1 to Warehouse 1
- x_{F2-DC}: Units shipped from Factory 2 to Distribution Center
- x_{DC-W2}: Units shipped from Distribution Center to Warehouse 2
- x_{W1-W2}: Units shipped from Warehouse 1 to Warehouse 2
- x_{W2-W1}: Units shipped from Warehouse 2 to Warehouse 1

The company wants to minimize total shipping cost:

Minimize $200x_{F1-F2} + 400x_{F1-DC} + 900x_{F1-W1} + 300x_{F2-DC} + 100x_{DC-W2} + 300x_{W1-W2} + 200x_{W2-W1}$

while considering the following constraints

(a) Production Constraints

$$\begin{aligned} \text{Factory 1 output:} \quad & x_{F1-F2} + x_{F1-DC} + x_{F1-W1} = 50 \\ \text{Factory 2 output:} \quad & x_{F2-DC} = 40 + x_{F1-F2} \end{aligned}$$

(b) Distribution Center Flow Conservation

$$x_{F1-DC} + x_{F2-DC} = x_{DC-W2}$$

(c) Warehouse Flow Conservation

$$\begin{aligned} \text{Warehouse 1 balance:} \quad & x_{F1-W1} + x_{W2-W1} - x_{W1-W2} = 30 \\ \text{Warehouse 2 balance:} \quad & x_{DC-W2} + x_{W1-W2} - x_{W2-W1} = 60 \end{aligned}$$

(d) Capacity Constraints

$$\begin{aligned} \text{F1} \to \text{F2 lane:} \quad & x_{F1-F2} \leq 10 \\ \text{DC} \to \text{W2 lane:} \quad & x_{DC-W2} \leq 80 \end{aligned}$$

(e) Non-negativity Constraints

$$x_{F1-F2}, x_{F1-DC}, x_{F1-W1}, x_{F2-DC}, x_{DC-W2}, x_{W1-W2}, x_{W2-W1} \geq 0$$

The optimal solution for the shipping problem is given by the following values:

$$\begin{gathered} x_{F1-F2} = 0, \quad x_{F1-DC} = 40, \quad x_{F1-W1} = 10, \\ x_{F2-DC} = 40, \quad x_{DC-W2} = 80, \\ x_{W1-W2} = 0, \quad x_{W2-W1} = 20. \end{gathered}$$

The resulting total shipping cost is $49,000.